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The Battle for Homes: How Does Home Sharing Disrupt Local Residential Markets?

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Abstract. As cities debate how to regulate Airbnb and other home-sharing services, we study the impacts of home sharing on local residential real estate markets. By accommodating transient travelers with short-term rental properties, home-sharing platforms have evolved as a major alternative channel that attracts the growing supply of residential properties. However, on the demand side of local residential markets, home-sharing platforms are not a viable option for residents. To demonstrate this dynamic between home sharing and local residential markets, we leverage a unique quasi-experiment on Airbnb—a platform policy that caps the number of properties a host can manage in a city—and find that the policy reduced rents (in the long-term rental markets) and home values (in the for-sale housing markets) by about 3% and did not affect the price-to-rent ratio. Consistent with the conjecture, we find that the policy impacts can be attributed to increased supply in local residential markets because of the policy. Quantitatively, our estimates suggest that, if the density of affected Airbnb properties is 1% higher in a market, the policy may further decrease rents and home values by about 0.03%–0.06%, which is similar across each policy-affected city. Our empirical findings add to the debate about the impacts of home sharing on local residential markets with a novel data set and a unique identification strategy. Practically speaking, our research is a timely response to the debate on regulating home sharing and has implications for various stakeholders of the residential real estate markets.

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1. Introduction

Home-sharing platforms allow individuals to earn extra income by opening their accommodation space to transient travelers.¹ The growth of home sharing, particularly the world’s largest platform Airbnb.com, has been exponential. Starting from renting out three air mattresses in 2008, Airbnb now hosts more than 6 million properties in nearly 100,000 cities across 191 countries (Airbnb 2019). A vital force of this growth is the significant interest from the supply side, such as absentee landlords who capitalize on these internet-based vacation rental platforms by potentially taking homes off local residential markets (Filippas et al. 2020). Criticism begins to arise that these Airbnb hosts with multiple properties cut off the supply in local markets, contributing to rising rents and home values (Sheppard and Udell 2016, Horn and Merante 2017, Barron et al. 2021).² In contrast, advocates insist that these hosts do not contribute to the housing

affordability concerns because whether their properties on Airbnb constitute a significant share of the local markets remains unclear (Stulberg 2016). Despite the controversy, legislators around the world are experimenting with policies to meet the goal of housing affordability while reaping the benefits of this “sharing economy.”³ To inform policy making, we examine in this study whether and how home sharing makes local rentals and/or housing less affordable.

Internet-based home sharing provides an alternative channel for homeowners, real estate agents, and other “suppliers” of residential properties. Two potential forces contribute to their interest in this alternative channel. On the one hand, the displacement of properties to home-sharing platforms incurs little costs for those owners. On the other hand, they may earn more than listing the properties in the traditional residential markets (i.e., long-term rental markets or for-sale housing markets). On the demand side, the traditional

residential markets primarily serve local residents. In contrast, on home-sharing platforms, because most property owners mainly provide short-term rental services (Barron et al. 2021), their customers are mostly transient travelers who typically spend only a few days in a locality. In other words, home sharing is not a viable option for local residents. We regard this unique relationship with local residential markets as an important feature of home sharing (i.e., as a major alternative of the local markets on the supply side but not on the demand side). Therefore, if the extent of the displacement (of residential properties from local markets to home sharing) is sufficiently significant, home sharing will make rental and housing markets less affordable. In this study, we examine whether home sharing has been a major alternative for the supply of residential properties.

We add to the debate about the impact of home sharing on residential real estate markets (Sheppard and Udell 2016, Horn and Merante 2017, Barron et al. 2021) by studying a unique quasi-experiment—a platform policy by Airbnb that caps the number of properties a host can manage in a city. Specifically, to ease the relationship with local governments, the so-called “One Host, One Home” policy (hereafter, the OHOH policy or the policy) requires a host to list properties only at a single address on Airbnb. Violations to this regulation may lead to the properties being removed by Airbnb and even the host accounts being suspended. Our empirical design relies on the staggered rollout of the policy, which was unexpectedly announced in April 2016 and then implemented in November 2016 for hosts in New York City and San Francisco. Later in February 2017, Airbnb implemented the same policy among hosts in Portland, Oregon (hereafter, Portland).⁴

The OHOH policy provides a unique chance to answer our research questions. To see this, the policy, on the one hand, forces Airbnb hosts to remove properties from the platform and, on the other hand, prevents them from displacing more properties from local markets.⁵ If home-sharing platforms had not been a major alternative for the supply in local markets, the policy would not have significantly affected (i.e., reduced) either rents or home values. In other words, economically meaningful impacts of the policy will serve as evidence of how home sharing disrupts local residential markets. Therefore, our empirical tests examine whether the OHOH policy reduced the rents and home values in local markets (and hence, improved housing affordability). If so, did the policy affect the price indices (i.e., rents and home values) by increasing the supply in local markets? We further quantify the policy impacts on market equilibrium outcomes and examine whether the estimates can be generalized to other cities or markets.

We combine a rich set of data from multiple sources, including a novel data set of all Airbnb listings in

the United States (obtained from AirDNA), Zillow and CoStar (for local residential markets), and local economic and demographic characteristics maintained by the U.S. Census Bureau. We construct our main sample at the zip code level (rather than at the city level). This granularity allows for better matching and sufficient variation across locations. Operationally, for zip codes in the three policy-affected cities (New York City, San Francisco, and Portland), we use a coarsened exact matching (CEM) method to construct a comparison group of zip codes from other U.S. cities. These “policy-unaffected” zip codes are similar to those affected in terms of Airbnb market size, local residential prices, and local economic and demographic characteristics but have not been subject to the OHOH policy.⁶ The sample spans a period from 2014 to 2017, covering a sufficiently wide window for the policy’s impacts in local residential markets to unveil.

Based on the CEM sample, we construct a zip code by month panel, and our main specification is standard differences-in-differences (DID) (Abadie 2005). In addition to zip code fixed effects and year-by-month fixed effects, we include city-specific time trends to control for unobserved heterogeneity in the overall trends in local markets. We also control a rich set of American Community Survey (ACS) variables, such as population, median household income, and unemployment rate. To further alleviate concerns regarding the comparability between policy-affected and unaffected markets, we apply a state-of-the-art synthetic control method that allows for multiple “treated” units (Xu 2017). Lastly, we rule out alternative explanations using a series of robustness checks and placebo tests. Section A1 of the online appendix summarizes all analyses in this paper.

Consistent with our conjecture, we first find that the OHOH policy reduced both rents and home values. Quantitatively, relative to unaffected markets, rents and home values in policy-affected zip codes decreased by 1.2% and 1.4%, respectively, after the announcement and further decreased by 1.9% and 1.5%, respectively, after the implementation. Taken together, both price indices in the policy-affected markets reduced by about 3%. In monetary terms, these price reductions translate to savings of about \$422 annually for a median renter and about \$7,057 in home values for a median buyer at the time of purchasing.⁷ As expected, the policy alleviated the housing affordability concerns in those affected zip codes. These economically significant impacts of the policy are consistent with our conjecture that home sharing has evolved as a major disruptor of local residential markets. In addition, consistent with the finding of similar impacts on rents and home values, we find an insignificant effect of the policy on the price-to-rent ratio (home values divided by annual rents).

The assumption for the validity of our empirical design is that the matched policy-affected and unaffected zip codes had relatively “parallel trends” in local residential markets before the policy. To verify the assumption (and also as a robustness check of our main findings), we estimate a relative time model in the sense of Angrist and Pischke (2008) and find that before the policy, the two groups of zip codes indeed had similar trends in both rental and housing markets. The reductions in rents and home values began to show up shortly after the announcement and continued to be significant after the implementation. Numerically, the reductions got larger in magnitude over time. These findings also demonstrate no expectations about whether, when, or where an OHOH-type policy would be rolled out; therefore, we argue that the OHOH policy can be considered a useful way to date the onset of local market equilibrium changes in response to policies adopted by home-sharing platforms.

Our second set of findings concerns the mechanism underlying the impacts of the OHOH policy (on local price indices). Because we argue that home sharing is a major disruptor of local markets on the supply side (but does not compete for local residents on the demand side), the OHOH policy should lead to increased supply in the local markets, which would, in turn, reduce local residential prices. We find that the policy indeed increased the supply in both rental and housing markets. Consistently, we find that the number of properties operated by “multi-listing” (ML) Airbnb hosts, who managed properties at different addresses and were hence subject to the policy restriction, shrank significantly after the policy. We also rule out several alternative explanations, such as the impact of demand and intervening events.

Third, although our empirical analyses build on the CEM process and control for an extensive set of factors, one may still worry that the estimated policy impacts do not apply to other markets. To overcome the challenge, we seek to quantify the intensity of the policy impacts by exploring the moderation of the density of policy-affected ML properties. Specifically, because the policy targets ML hosts, we use the (logged) number of their properties per 1,000 population as the intensity measure and examine how it moderates the policy impacts. Our estimates suggest that the policy may further decrease home values and rents by about 0.03%–0.06% if the density is 1% higher in a zip code. For a median renter in the policy-affected zip codes in our sample, the estimates suggest that the policy may help the household save an extra \$75 a year if the density of ML properties is 10% higher. Similarly, based on the Zillow Home Value Index (ZHVI) in the affected zip codes, the extra saving for a home buyer is about \$803 if the density of ML properties is 10% higher. Consistent with the

impacts on rents and home values, we also find that the policy would further increase the supply in local residential markets if the density of affected properties was higher.

We further contemplate the generalizability of our results by evaluating the policy impacts and the intensity for each affected city separately. For each city, we reconstruct a sample of matched zip codes by replicating the CEM process (i.e., looking for matched zip codes across the country). Across the three cities, we find that the policy reduced both rents and home values. In terms of magnitude, the policy had the largest impacts on zip codes in San Francisco, followed by Portland and then, New York City. We attribute this difference in the policy impacts to the difference in the density of affected properties. Compared with New York City and Portland, the zip codes in San Francisco had the highest density of ML properties in our study period.⁸ We account for this discrepancy and replicate our intensity analysis for each affected city. We find that across the three cities, the intensity of the policy impacts is quantitatively close to each other.

Our empirical findings have important implications. The growing popularity of home sharing has invited debates about regulating such platforms (Filippas and Horton 2017). Only if we come to a better understanding of home sharing’s relationship with local residential markets can the legislators, the city administration, and the marketplace itself better capitalize on this “sharing economy.” Our empirical evidence demonstrates that home sharing has evolved as a major alternative to the local residential markets on the supply side. The traditional channels are long-term rental markets and for-sale housing markets. Our work suggests that home sharing has evolved as the third major option for property owners. However, on the demand side, home sharing may not be a viable option for local residents (in addition to the traditional rental and housing markets). In our opinion, this “asymmetry” in the relationship between home sharing and local residential markets speaks directly to the rising concern of housing affordability across the United States and particularly, in major metropolitan areas (Metcalf 2018). Furthermore, our estimates of the policy impacts and the intensity provide a useful framework and reference points for policy makers to evaluate to what extent introducing OHOH-type restrictions may mitigate their local affordability concerns.

The remainder of the paper is organized as follows. We begin with a summary of the related literature and highlight our contributions in the next section. Section 3 introduces and discusses the OHOH policy. Section 4 describes the data, how we construct the main sample and various measures, and our empirical strategies. We report our findings in Section 5. Section 6 concludes the

paper. Lastly, we present the results from a plethora of additional tests in the online appendix.

2. Literature and Contributions

2.1. Literature on Home Sharing

Studies on the impacts of home sharing are closest in spirit to ours. Early work focuses on how home sharing affects the incumbents—the short-term accommodation providers (e.g., hotels)—in a competitive landscape (Zervas et al. 2017, Farronato and Fradkin 2018, Li and Srinivasan 2019). More recently, a few studies have documented the emergence of home-sharing services and rising prices in local residential markets (Sheppard and Udell 2016, Horn and Merante 2017, Barron et al. 2021). We contribute to this literature in a few fundamental ways. First, our empirical findings rely on a unique quasi-experiment—the OHOH policy—to demonstrate the relationship between home sharing and local real estate markets. This research agenda is a useful way to date the onset of changes in local market equilibrium in response to platform-mandated policies, like the OHOH restriction. Second, our findings differ from the existing literature (especially Barron et al. 2021). We find that the OHOH policy had quantitatively similar impacts on rents and home values, which adds to the debate about how home-sharing services affect local residential markets. Last but not least, our estimates of the policy impacts and the intensity provide a framework for policy makers and legislators to evaluate the impacts of home sharing in their localities.

Within this strand of literature, our findings parallel those of Barron et al. (2021). They show that a 1% increase in Airbnb listings in a market is associated with an increase of \$108 in annual rents and \$1,800 in home values. Although both studies examine the relation between home-sharing services and local real estate markets, we differ from them in the following ways. First, our findings hinge on the unique quasi-experiment—the OHOH policy—and are deployed through a handful of tests by devising different empirical methods. Barron et al. (2021), instead, use an instrumental variable approach based on the search interest of Airbnb on Google (Goldsmith-Pinkham et al. 2020).⁹ Second, we report that the OHOH policy affected the rental and housing markets relatively equally (i.e., similar reductions in rents and home values and an insignificant impact on the price-to-rent ratio). We attribute our findings to the general equilibrium in real estate markets that the relative value of home values with respect to rents stays largely constant in the long run (Sommer and Sullivan 2018). Barron et al. (2021), however, find that Airbnb's impact in housing markets is slightly larger than that in long-term rental markets. They argue that home

sharing increases the option value of residential properties. Lastly, our data come from a leading data analytics company that focuses on home-sharing services. The main advantages are that the data set not only covers all cities (zip codes) in the United States but also, records both consumer-facing information (such as property characteristics, like room type) and backstage information (property availability, etc.). Instead, Barron et al. (2021) collect their data from scraping Airbnb listing pages.

With the differences summarized above, we view our study as a complement of Barron et al. (2021) on how home-sharing services, such as Airbnb, affect local residential real estate markets. Our findings from the unique quasi-experiment (the OHOH policy) add to the confidence that home sharing has evolved as a major alternative for the supply of residential properties, leading to concerns about the shortage in local housing and long-term rental markets.

2.2. Economics of Real Estate Markets

Our work is also related to the extensive literature on the economics of residential real estate market, one of the most important segments in any economy. In the United States, as an example, the full stock of the residential real estate market reached \$36.2 trillion in 2020 (Richardson 2021). The residential markets are often characterized by some unique features. For example, it is well recognized that, first, the supply in residential markets is often constrained because of limited land space and regulatory restrictions (Glaeser and Gyourko 2018). Second, the demand for residential housing (either long-term rentals or for-sale houses) is, in general, inelastic with respect to income and price (Mayo 1981). A long-lasting series of studies has focused on estimating the (income and price) elasticity of housing demand (see, e.g., Davis and Ortalo-Magné 2011 and Albouy et al. 2016 for more recent estimates).

We add to our understanding of the residential real estate markets by presenting evidence that internet-based home sharing has evolved as another major channel for homeowners, real estate agents, and other suppliers of residential housing to list their properties. The traditional channels are primarily long-term rental markets and for-sale housing markets. Our work demonstrates that home sharing has evolved as the third major option for property owners, especially in large metropolitan areas. Our findings also illustrate the new interactions between home sharing and the traditional residential markets. That is, along with other studies on the impacts of home sharing on local residential markets, our empirical findings suggest that if we overlook this alternative channel for home supply, future theoretical or empirical studies would

lead to a biased understanding of the market as a whole.

In addition, our empirical findings that the OHOH policy had similar impacts on rents and home values echo a consensus in the literature that under relatively mild conditions, in the long run, the equilibrium home values equal the expected discounted rents (de Leeuw and Ekanem 1971, Eubank and Sirmans 1979, Rosen and Smith 1983, Smith et al. 1988, Wheaton 1990, Sommer et al. 2013, Sommer and Sullivan 2018). In other words, the relative value of home values and rents is largely stable in the long term. Consistent with the prediction, our estimates of the policy impacts and the intensity of the policy impacts all suggest that the policy reduced rents and home values in similar magnitudes, which is supported by the insignificant impacts on the price-to-rent ratio (i.e., the ratio between the home value and annual rent (or $12 \times$ monthly rent)).

2.3. Other Related Literature

Our work contributes to the emerging literature on peer-to-peer (P2P) markets in general (Einav et al. 2016, Hall and Krueger 2018, Hui et al. 2016, Fradkin 2017, Li and Netessine 2019, Filippas et al. 2020). As technology-infused disruptors, P2P platforms are characterized by sufficiently low transaction cost (Zervas et al. 2017) and relatively low entry barriers (Einav et al. 2016), through which suppliers with underutilized resources (space, time, money, etc.) can meet on-demand needs flexibly. Additionally, P2P platforms provide enhanced reach to customers by reducing search costs (Einav et al. 2016) and facilitating arm's length transactions (Edelman et al. 2017). We add to this literature by studying a unique P2P market—home-sharing services. Our empirical evidence points out a unique relationship between home sharing and the traditional residential markets. That is, home sharing is a substitute for the supply in local residential markets but does not compete for local residents on the demand side.

More broadly, we are also related to the literature studying the dynamics between online and offline markets (Forman et al. 2009, Overby and Forman 2015). The literature has explored a wide variety of markets, such as retailing and consumer goods (Brynjolfsson et al. 2009), banking (Abhishek et al. 2019), ride-hailing services (Greenwood and Wattal 2017, Babar and Burtch 2020), and marketplace lending (Alyakoob et al. 2021). We are among the first to explore the relationship between online home sharing and local residential markets, which has not been explored extensively in the literature. We argue that understanding the relationship is fundamentally important because the residential real estate markets are large in volume (Richardson 2021) and impact

many aspects of an economy. Our findings demonstrate that in addition to the traditional rental and for-sale housing markets, internet-based home sharing has evolved as a major option for residential property owners.

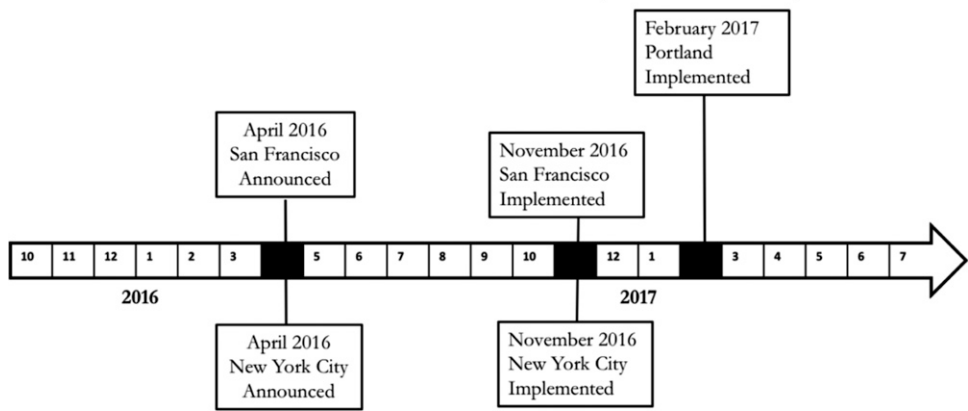
3. Airbnb's "One Host, One Home" Policy

Airbnb is known to provide travelers with authentic experiences by allowing them to "live like a local" (Benner 2016). In the early years, inviting travelers to stay in their spare space while making some extra money was the primary motivation of most hosts. More recently, some hosts have realized that Airbnb can be a profit generator by renting out a second, third, or even more than 1,000 homes on the platform. Although these hosts help scale up the platform, they have displaced many local residential properties to home-sharing platforms.¹⁰ Fearing the potential harm to local residential markets, many cities in the United States have been engaged in court battles with Airbnb (Carville et al. 2020). Despite its reluctance to cooperate with local governments, between 2016 and 2017, Airbnb rolled out the "One Host, One Home" policy aiming to promote home sharing in primary residences only. The policy restricts hosts in San Francisco, New York City, and Portland from listing properties at multiple addresses. Such an unexpected move by Airbnb was considered as a sign to smooth the relationship with local governments and city legislators (Lin 2017).

Figure 1 depicts the time line of the policy rollouts in the three cities. Specifically, the OHOH policy was unexpectedly announced on April 2, 2016 when Airbnb publicized its commitment to promoting home sharing in primary residences only by "working with leaders in New York City and San Francisco on progressive policies" that limit "hosts to sharing listings at just one address" on the platform (Airbnb 2016, 2017a). Seven months later, the policy was officially implemented. Effective on November 1, 2016, hosts in New York City and San Francisco could list properties at only one address. If a host wanted to list a property from a different address, she would be forced to remove the existing properties. Violating the policy may lead to her properties and Airbnb account being suspended or even deactivated.¹¹ Subsequently, on January 30, 2017, Airbnb launched the same policy in Portland to address "Portland's housing affordability crisis and unwanted commercial operators who may be converting housing to illegal hotels" on the platform (Airbnb 2017b, p. 1). The policy was immediately effective in Portland.

The market participants, including hosts and guests, were largely uncertain about the timing, the locations, or the format of the policy.¹² This unexpected, staggered

Figure 1. Timeline of the “One Host, One Home” Policy



Notes. The months when the policy was announced or implemented are indicated in black. Because the policy was implemented in Portland on January 30, 2017 (end of the month), we mark February as the actual month of policy implementation in Portland.

rollout of the policy in a selected number of cities can be viewed as a shock to the participation of Airbnb hosts. Specifically, the policy imposes restrictions on these Airbnb hosts’ participation on the platform and essentially attenuates the trade-off between listing properties in home sharing and local residential markets. Therefore, the OHOH policy offers a unique chance to study the dynamics between home-sharing and local residential markets.

4. Data and Methods

4.1. Data

We collect data from multiple sources: the short-term rental data analytics company AirDNA, real estate information companies Zillow and CoStar, and the U.S. Census Bureau.

First, our property-level data for Airbnb come from AirDNA, which provides trusted data and analytic services about short-term rental markets. AirDNA’s data service has been widely used and endorsed by leading industry clients, such as VACASA, the CBRE Group, the Los Angeles Tourism & Convention Board, and Bank of America Merrill Lynch.¹³ The data set contains a monthly panel of all individual properties in the United States. Our observation period is between 2014 and 2017 (a total of 48 months), which covers 27 months before and 21 months after the announcement of the policy (in April 2016). We choose this relatively large time window for the policy impacts to unravel fully. An advantage of the AirDNA data set, compared with the web-scraped data used in other studies (e.g., Li et al. 2016), is that it contains both consumer-facing information (such as property characteristics, like room type) and backstage information (e.g., property availability, booking information (including nightly rate and number of reservations), and the unique geo-coordinates of each property).¹⁴ In the data set, a property can be marked unavailable for booking because

the host has blocked the calendar or other guests have booked the property. By observing both availability and booking information, we can effectively distinguish between being blocked and being booked. We follow Li and Srinivasan (2019) to define active listings as the properties marked as either booked or available on the calendar. We then use this information to calculate the number of properties operated by a host and categorize hosts into ML hosts and others. We also refer to properties operated by ML hosts as ML properties.

Second, we collect data on local housing and rental markets from two real estate information companies: Zillow and CoStar. As the largest online residential real estate platform, Zillow aggregates the median estimated market rents and home values into smoothed, seasonally adjusted measures, namely Zillow Rent Index (ZRI) and Zillow Home Value Index (ZHVI), for each zip code in each month.¹⁵ Based on ZRI and ZHVI, we calculate the price-to-rent ratio (Price2Rent hereafter), which is the home value divided by annual rent (i.e., $ZHVI / (12 \times ZRI)$). Zillow also publishes the monthly supply of for-sale housing in a zip code in a month. We complement the Zillow data set with the quarterly rental supply provided by CoStar. The data on housing and rental supply allow us to examine the mechanisms behind the policy impacts.

Lastly, because the economic and demographic characteristics of a market will arguably play a role in determining rents and home values, we add an extensive set of annually compiled control variables in each zip code from the U.S. Census Bureau’s ACS. The characteristics include population; number of households; median household income; unemployment rate; home vacancy rate; and fractions of the White, Hispanic, and female population.

We match these four sources of data at the zip code by month level. We exclude zip codes without Airbnb

presence or local price indices because of data unavailability. The full sample consists of 3,274,129 Airbnb properties listed in roughly 200 zip codes of the three policy-affected cities and 13,164 zip codes from other U.S. cities.

4.2. Empirical Design and Sample Construction

Capitalizing on the OHOH policy, our empirical strategy is a usual DID design combined with matching (Table 1). Zip codes in the three policy-affected cities form the “treated” group, which has been under policy restrictions. In contrast, we use a CEM method to construct a “control” group of zip codes that had not been subject to the OHOH policy by the end of our study period. The matching method has been widely adopted in the literature (see, e.g., Iacus et al. 2011 and Overby and Forman 2015).

In our CEM method, we first separate affected and unaffected zip codes into different strata by “coarsening” a set of zip code-level variables into bins. These variables include the Airbnb market size, the local residential prices (ZRI and ZHVI), and the characteristics of the local economy and demographics. We then apply an “exact matching” algorithm to the coarsened data to determine the matches. In essence, the algorithm compares affected and unaffected zip codes assigned to the same strata in which all matching covariates, namely the observed characteristics, are similar. We retain only the strata that have at least one affected zip code and one unaffected zip code (Iacus et al. 2012). The result is a matched sample of 87 zip codes in the affected cities and 87 zip codes from other cities. In total, there are 174 zip codes in the CEM-generated sample.

Table 2 presents the definitions and summary statistics of the key variables for the CEM sample. We translate the rents and home values—ZRI and ZHVI, both in \$1,000—into 1990 U.S. dollars using the historical inflation rate.¹⁶ It should be noted that the price-to-rent ratio remains the same despite the translation. In our empirical analyses, we take logarithms of these three price indices. As such, our findings inherit the semi-elasticity interpretation. Additionally, we take logarithms of several control variables that exhibit skewed distributions. In regressions, we winsorize the variables at the 1% and 99% levels to control for potential bias because of extreme values.

Table 1. The Differences-in-Differences Design

Group	Before policy	After policy
Zip codes in policy-affected cities	O	X
Unaffected zip codes	O	O

Note. “Policy” indicates the rollout of the OHOH policy, which includes both announcement and implementation.

One may worry that the affected cities, especially New York City and San Francisco, are different from other U.S. cities. First, it should be noted that our empirical tests are based on a zip code by month panel. Although the affected cities may look different, using our CEM method, it is not surprising to find highly comparable zip codes in the country that can mirror some affected ones (in terms of the matching covariates). Second, in the matching process, some zip codes in the affected cities cannot be grouped in any matched strata and are, hence, dropped. Figure 2 labels the matched zip codes in the three affected cities. Note that the “expensive” zip codes in New York City (in terms of rents or home values), particularly those in the Manhattan area, cannot be matched and are dropped. Similarly, the “expensive” areas in San Francisco, such as Seacliff, Nob Hill, and the Financial District, are not included in the CEM sample either. In other words, our CEM method helps filter out some “unique” areas in the affected cities and does not include them in our analyses. In contrast, as the figure illustrates, most zip codes in Portland can be matched and are included in our analysis.

Table 3 reports balance checks that compare the affected and unaffected zip codes in the CEM sample on observable characteristics evaluated using observations before the policy. We conduct paired *t* tests. One can see that, on average, the policy-affected zip codes were not significantly different from the unaffected ones before the OHOH policy.

4.3. Main Regression Specification

Using the CEM sample, our main empirical model is a DID regression. We separate the announcement from the implementation because announcing the policy might have already changed the hosts’ expectations and their property allocation. For each zip code *i* in month *t*, the policy impacts on some market outcomes are specified as

$$Y_{it} = \beta_1 1(\text{Announcement})_{it} + \beta_2 1(\text{Implementation})_{it} + \gamma' Z_{it} + \mu_t + v_i + \varepsilon_{it}, \quad (1)$$

where the indicator $1(\text{Announcement})_{it}$ equals one if the policy has been announced in zip code *i* by month *t*; $1(\text{Implementation})_{it}$ equals one if the policy has been implemented in zip code *i* by month *t*; Z_{it} is a vector of control variables, including economic and demographic characteristics (ACS controls) in each zip code and city-specific time trends (i.e., for each city, we construct an ordered index whose value indicates a month in our study period); μ_t denotes year-by-month fixed effects; v_i denotes zip code fixed effects; and ε_{it} is the idiosyncratic error term. Depending on the specific test, the dependent variable Y_{it} can be rents, home values, or other measures such as the supply in local

Table 2. Variable Definitions, Data Sources, and Summary Statistics (from the CEM Sample)

Source	Variable	Definition	Mean	Median	Standard deviation	Min	Max
Rental and housing market price indices Zillow	ZRI	ZRI in \$1,000 (in 1990 US dollars)	1.23	1.16	0.37	0.62	2.63
	ZHVI	ZHVI in \$1,000 (in 1990 US dollars)	282.14	249.20	143.85	73.05	860.44
	Price2Rent	ZHVI/(12 × ZRI)	18.36	17.65	4.46	9.17	32.12
The OHOH policy Airbnb	1(Announcement)	A dummy indicating the policy announcement	0.16	0	0.36	0	1
	1(Implementation)	A dummy indicating the policy implementation	0.14	0	0.34	0	1
Supply on Airbnb AirDNA	Airbnb properties	The number of Airbnb properties	84.14	34	129.24	1	816
	ML properties	The number of Airbnb properties operated by multilisting hosts	13.97	3	27.89	0	172
	New properties	The number of new properties on Airbnb	9.28	4	16.23	0	104
Supply in the local residential markets Zillow	Housing supply	The number of for-sale houses	122.64	103	75.56	24	412
	Rental supply	The number of available rental units (at the quarterly level)	672.21	275	1,053.70	0	5405
Economic and demographic controls U.S. Census Bureau: The American Community Survey	Population	Population (in 1,000)	36.59	34.57	19.91	0.86	106.60
	Households	The number of households (in 1,000)	13.14	12.51	6.63	0.40	35.00
	Med. income	The median household income (in 1,000)	68.43	64.20	21.71	26.76	161.42
	Unemployment rate	Unemployment rate (%)	7.56	7.20	2.48	2.60	14.40
	Home vacancy rate	Home vacancy rate (%)	7.32	6.37	5.31	2.42	47.69
	% White	Percentage of the White population	41.78	42.79	22.50	1.05	81.71
	% Hispanic	Percentage of the Hispanic population	16.54	12.50	12.74	1.01	56.14
	% Female	Percentage of the female population	51.29	51.20	1.55	45.45	55.31

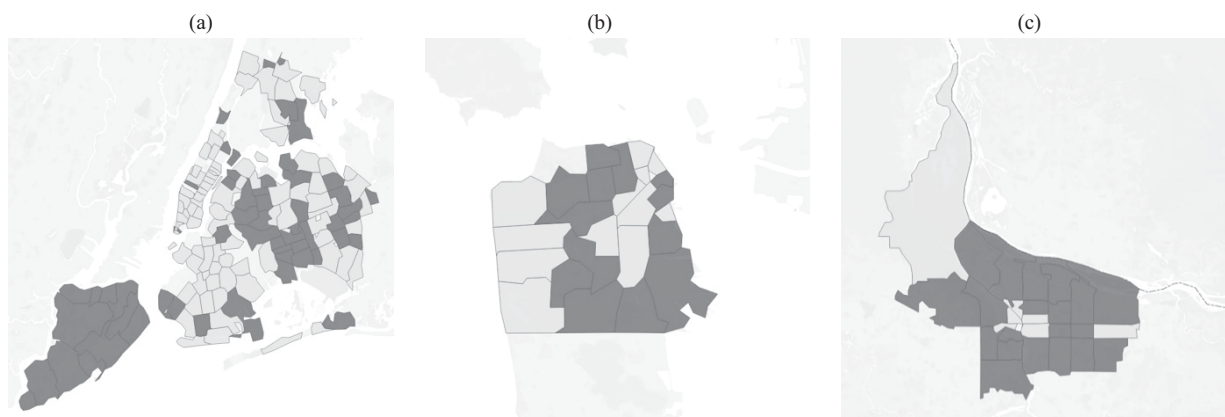
residential markets. In regressions, we cluster the standard error at the zip code level to account for potential serial correlations (Bertrand et al. 2004).

The coefficients β_1 and β_2 capture the policy impacts. It is worth pointing out that the corresponding variables, $1(\text{Announcement})_{it}$ and $1(\text{Implementation})_{it}$, are both time and location specific. Taking $1(\text{Implementation})_{it}$ as an example, it equals one if zip code i is in New York City or San Francisco and month t is after November 2016. It is also one if zip code i is in Portland

and month t is after January 2017. These two variables take values of zeros for all unaffected zip codes.¹⁷

5. Findings

This section reports our empirical findings. As a quick road map, we first present formal estimates from the main DID specification (Equation (1)). Following that, we drill down to the mechanisms underlying the policy impacts by showing how the policy increased the

Figure 2. Matched Zip Codes (in Dark Gray) in the CEM Sample

Notes. (a) New York City. (b) San Francisco. (c) Portland.

Table 3. Characteristics of Unaffected Zip Codes vs. Affected Zip Codes in the CEM Sample

	Unaffected zip codes			Affected zip codes			<i>t</i> test	
	Mean	Standard deviation	<i>N</i>	Mean	Standard deviation	<i>N</i>	Difference	<i>p</i> -value
Rental and housing market price indices								
log(<i>ZRI</i>)	0.158	0.267	87	0.120	0.279	87	−0.038	0.358
log(<i>ZHVI</i>)	5.469	0.470	87	5.477	0.470	87	0.008	0.906
log(<i>Price2Rent</i>)	2.826	0.238	87	2.873	0.246	87	0.047	0.205
The market size of Airbnb								
log(<i>Airbnb Properties</i>)	3.093	1.321	87	3.290	1.504	87	0.197	0.364
Economic and demographic characteristics								
log(<i>Population</i>)	10.302	0.605	87	10.368	0.677	87	0.066	0.499
log(<i>Households</i>)	9.305	0.570	87	9.363	0.631	87	0.058	0.526
log(<i>Med. income</i>)	11.077	0.295	87	11.040	0.298	87	−0.037	0.413
Unemployment rate	8.099	2.418	87	8.496	2.436	87	0.397	0.283
Home vacancy rate	7.040	5.353	87	7.716	6.110	87	0.676	0.438
% White	42.222	22.061	87	41.867	23.319	87	−0.356	0.918
% Hispanic	17.058	13.246	87	15.688	12.248	87	−1.370	0.480
% Female	51.296	1.602	87	51.405	1.654	87	0.109	0.659

supply in local residential markets. Then, we quantify the intensity of the policy impacts and conduct by-city analyses for generalizability and policy implications. Lastly, we provide a battery of tests to verify the robustness of our main results and rule out alternative explanations.

5.1. Policy Impacts on Local Residential Market Prices

5.1.1. Results from the DID Analysis. Table 4 reports our main findings of the policy impacts on price indices per Equation (1). We omit the coefficients for city-specific time trends because of space constraints. Column (1) shows that rents, on average, decreased by 1.2% after the announcement and another 1.9% after the implementation in affected zip codes relative to unaffected zip codes. In total, the rents decreased by 3.1% in policy-affected zip codes, which translates to a saving of about \$422 annually ($\approx$ median rent \$1,134.6 $\times$ 3.1% $\times$ 12 months) for a typical renter household. Similarly, estimates in column (2) reveal that the home values, on average, decreased by 1.4% after the announcement and another 1.5% after the implementation.¹⁸ Overall, home values decreased by about 2.9%, an equivalent of paying \$7,057 less ($\approx$ median home value \$243,328.7 $\times$ 2.9%) at the time of purchase for a typical home buyer in the policy-affected zip codes.

Notably, the magnitudes of the overall policy impacts reported in columns (1) and (2) are similar, suggesting that the policy had similar impacts on rents and home values. Consistently, the estimated impact on the price-to-rent ratio was not statistically significant, as in column (3). In other words, the value of rents relative to home values stayed constant before and after the policy. We also find some interesting patterns in the relationships between the price indices

and some characteristics of the local economy and demographics. For example, rents are, on average, lower in zip codes with more households. The income level is positively correlated with rents and home values; in contrast, the unemployment rate has a negative relationship with both rents and home values. The higher the housing vacancy rate in a zip code is, the lower the prices are.

5.1.2. Evidence from a Relative Time Model. An important assumption for the validity of our DID design (per Equation (1)) is that the affected and unaffected zip codes had relatively “parallel trends” in local residential markets before the policy. Estimating a relative time model in the sense of Angrist and Pischke (2008) allows us to check the assumption. In addition, results from the relative time model can demonstrate how the policy impacts evolve over time. We specify our relative time model as

$$Y_{it} = \sum_{\tau=-T}^N \lambda_{\tau} D_{i\tau} + \gamma' Z_{it} + \mu_t + \nu_i + \varepsilon_{it}, \quad (2)$$

where $D_{i\tau}$ is a dummy variable indicating whether month t is the τ -th month before (for negative τ) or after (for positive τ) the policy announcement in April 2016. Equation (2) is similar to Equation (1), except the two policy dummies are replaced by a set of dummy variables $D_{i\tau}$, each indicating a month. Note that these dummy variables are equal to zero for all unaffected zip codes. The set of coefficients λ_{τ} helps us determine whether there exists a similar prepolicy trend in the affected and unaffected zip codes and how the policy impacts evolve over time.

Using the CEM sample, we estimate Equation (2) (for rents, home values, and the price-to-rent ratio) with $T = 6$ and $n = 11$ to inspect the trends from six months before the announcement and a year after. We

Table 4. Policy Impacts on Local Residential Market Prices

	(1) log(ZRI)	(2) log(ZHVI)	(3) log(Price2Rent)
1(Announcement) _{it}	−0.012** (0.005)	−0.014** (0.006)	−0.002 (0.007)
1(Implementation) _{it}	−0.019*** (0.007)	−0.015*** (0.005)	0.004 (0.006)
ACS controls: Economic and demographic characteristics			
log(Population)	0.001 (0.145)	−0.104 (0.138)	−0.105 (0.173)
log(Households)	−0.335 (0.214)	−0.026 (0.197)	0.309 (0.211)
log(Med. income)	0.134* (0.075)	0.087 (0.072)	−0.048 (0.066)
Unemployment rate	−0.001 (0.003)	−0.005* (0.003)	−0.004 (0.003)
Housing vacancy rate	−0.006* (0.003)	−0.002 (0.003)	0.004 (0.003)
% White population	0.001 (0.002)	0.003 (0.002)	0.002 (0.002)
% Hispanic population	0.004 (0.003)	0.000 (0.003)	−0.004 (0.003)
% Female population	0.002 (0.003)	0.000 (0.003)	−0.001 (0.003)
City-specific time trends	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes
Observations	8,352	8,352	8,352
Adjusted R ²	0.987	0.997	0.978

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

also include a dummy for all months at least 7 months before the announcement and another dummy to indicate all months at least 12 months after. Because of the collinearity constraint, we drop the month immediately before the announcement (March 2016) and use it as the benchmark. We do not include the zip codes in Portland in this analysis as Portland implemented the policy in February 2017 without announcement. All other control variables (i.e., ACS characteristics, city-specific time trends, zip code fixed effects, and year-by-month fixed effects) remain.

We report the results in Table 5. The insignificant coefficients of time dummies before the announcement ($D_{\tau}, \tau = -6^+, -6, \dots, -1$) in columns (1) and (2) lend strong support to the parallel trend assumption.¹⁹ These results suggest that there were no significant prepolicy differences in the trends in local residential markets. After the announcement (D_0), the coefficients of time dummies become negative and statistically significant, suggesting that rents and home values started dropping in policy-affected zip codes (relative to unaffected areas). Both price indices continued to drop after the implementation (D_7). Additionally, the magnitude of the time dummy coefficients gets larger over time (i.e., more negative),

suggesting that the policy impacts indeed unravel as time passes. Lastly, results reported in column (3) show that the price-to-rent ratio stayed relatively constant over time between affected and unaffected zip codes. All these findings are qualitatively consistent with our main findings (Table 4).

5.2. Mechanisms Underlying the Policy Impacts on Local Residential Market Prices

We have shown that the policy led to a drop in both rents and home values. In this section, we seek to establish the mechanisms underlying the policy impacts. We also rule out several alternative explanations.

5.2.1. Policy Impacts on Supply in Local Residential Markets.

As discussed in Section 1, by restricting the participation of hosts who manage properties at multiple addresses on Airbnb, the OHOH policy may, on the one hand, remove properties from the platform and force the hosts to list them in the local residential markets. On the other hand, the policy may effectively prevent hosts from displacing more properties from the local residential markets to Airbnb. Taken together, we hypothesize that the policy infused

Table 5. Evidence from a Relative Time Model

	(1) log(ZRI)	(2) log(ZHVI)	(3) log(Price2Rent)
D ₋₆₊	-0.010* (0.006)	0.005 (0.004)	0.014** (0.006)
D ₋₆	-0.004 (0.007)	-0.001 (0.006)	0.003 (0.007)
D ₋₅	0.000 (0.008)	-0.001 (0.006)	-0.001 (0.007)
D ₋₄	0.003 (0.008)	-0.002 (0.005)	-0.004 (0.007)
D ₋₃	0.004 (0.007)	-0.002 (0.005)	-0.006 (0.007)
D ₋₂	0.004 (0.007)	-0.000 (0.005)	-0.004 (0.007)
D ₀ (Announcement)	-0.006 (0.007)	-0.002 (0.005)	0.004 (0.006)
D ₁	-0.012* (0.007)	-0.004 (0.005)	0.008 (0.006)
D ₂	-0.016** (0.007)	-0.007 (0.005)	0.009 (0.006)
D ₃	-0.018*** (0.006)	-0.012** (0.005)	0.006 (0.006)
D ₄	-0.022*** (0.006)	-0.015*** (0.005)	0.007 (0.007)
D ₅	-0.025*** (0.006)	-0.018*** (0.005)	0.007 (0.007)
D ₆	-0.025*** (0.006)	-0.019*** (0.005)	0.006 (0.007)
D ₇ (Implementation)	-0.024*** (0.006)	-0.018*** (0.006)	0.006 (0.007)
D ₈	-0.023*** (0.006)	-0.016*** (0.006)	0.007 (0.008)
D ₉	-0.021*** (0.006)	-0.015** (0.006)	0.006 (0.008)
D ₁₀	-0.021*** (0.007)	-0.018*** (0.006)	0.003 (0.008)
D ₁₁	-0.024*** (0.007)	-0.019*** (0.006)	0.005 (0.008)
D ₁₂₊	-0.041*** (0.006)	-0.021*** (0.005)	0.020*** (0.006)
ACS controls	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes
Observations	7,152	7,152	7,152
Adjusted R ²	0.988	0.997	0.978

Notes. In this table, we estimate Equation (2) excluding zip codes in Portland because it had only the policy implementation and a different time line. Robust standard errors clustered at the zip code level are shown in parentheses. We omit the coefficients of ACS controls in this table and all subsequent tables. The full tables (with ACS control coefficients) are reported in Section A4 of the online appendix. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 6. Policy Impacts on the Supply in Local Housing and Rental Markets

	(1) log(Housing supply)	(2) log(Rental supply)
1(After policy) _{it}	0.111*** (0.015)	0.024*** (0.009)
ACS controls	Yes	Yes
City-specific time trends	Yes	Yes
Zip code FE	Yes	Yes
Year-by-month FE	Yes	Yes
Observations	4,992	8,352
Adjusted R ²	0.501	0.235

Notes. Robust standard errors clustered at the geographic area level are shown in parentheses. For rental market supply, because the two policy dummies are highly correlated in our quarter-level data, we combine them into one variable 1(After Policy). We define 1(After policy) to be one if 1(Announcement) equals one in New York City and San Francisco or if 1(Implementation) equals one for Portland. The sample spans 2014 to 2017. The results are qualitatively similar if we use 1(Announcement) and 1(Implementation) separately. The zip code-level ACS controls are also aggregated to the level of geographic areas. FE stands for fixed effect.

*** $p < 0.01$.

supply into local residential markets, which in turn, led to decreases in rents and home values.

Table 6 reports the results using a similar regression with our main DID specification as Equation (1). Instead of having the announcement and implementation dummies separately, we construct a combined policy dummy, 1(After policy), which equals one after the announcement of the policy for the zip codes in New York City and San Francisco or after February 2017 (included) for the zip codes in Portland. We estimate this slightly different specification because our rental supply data are at a less granular level (quarterly). Because of the concern that the two policy dummies are highly correlated at the quarterly level, we combine them into one variable, 1(After Policy), in the regression of rental supply (column (2)). To be consistent, we estimate a similar specification for the regression of housing supply (column (1)). Nevertheless, the results are qualitatively the same if we estimate the policy impacts on housing supply using two policy dummies with Equation (1).

Column (1) of Table 6 reports the policy impact on housing supply. The dependent variable is log(*Housing supply*), the logged number of for-sale housing units listed in a zip code in a particular month. Note that we have fewer observations than the sample we use to study the policy impacts on prices because of the unavailability of supply data in some areas. The estimates show that the number of for-sale houses increased by about 11% after the policy. Again, this result controls the time trends at the city level, the zip code fixed effects, the year-by-month fixed effects,

and all observed characteristics of the local economy and demographics. To put the estimates into perspective, the median number of housing units listed for sale in a zip code in the policy-affected cities was 102 during our study period. Therefore, our estimates suggest that the policy added about 11 more for-sale listings ($\approx 102 \times 11\%$) to the housing market per zip code per month.

Column (2) of Table 6 shows the policy impact on the supply of rental properties. As pointed out in the last section, we obtain the rental supply information from CoStar. The CoStar supply data are available at the zip code by quarter level, which is less granular than our main sample (at the monthly level). Hence, we merge the quarterly rental supply data and our main sample to estimate the policy impact. The estimate suggests that the rental supply increased by about 2.4% after the policy. This result is consistent with our hypothesis (that the policy infused supply into local residential markets) and lends further support to the conjectured mechanism. Numerically, the quarterly median number of available rental units in a policy-affected zip code was about 340 in our sample. Hence, the 2.4% increase in the supply implies that about eight more rental units ($\approx 340 \times 2.4\%$) were added to a geographic area per month after the policy.

5.2.2. Policy Impacts on the Supply on Airbnb. The increased supply in local residential markets may come from removed properties from Airbnb or the prevented potential entrants to Airbnb from local residential markets. We, therefore, investigate the policy impacts on these two aspects of the supply on Airbnb. On the one hand, by restricting the participation of ML hosts, we expect that the policy would effectively reduce the supply from these hosts on the platform. Column (1) of Table 7 reports the coefficients of the two policy dummies from the regression, with the (logged) number of properties managed by ML hosts

being the dependent variable. The statistically significant and negative estimates suggest that the policy indeed reduced their supply on Airbnb. Quantitatively, the supply from ML hosts decreased by about 38% after the policy.

On the other hand, relative to unaffected zip codes, we expect the number of new properties (i.e., entrants) on Airbnb to decrease after the policy. Column (2) of Table 7 reports the coefficients of the policy dummies from the regression, with the (logged) number of newly listed properties for each zip code in each month being the dependent variable. Although not statistically significant, the negative coefficients suggest that the policy caused less entry into the platform.²⁰ Quantitatively, the number of entrants decreased by about 11% after the policy. These results on the supply on Airbnb validate the underlying mechanism of the policy impacts (i.e., causing the displacement of residential properties from the online channel to local residential markets).

5.2.3. Ruling out Alternative Explanations. Several alternative explanations for the decline in rents and home values merit careful study. This section investigates these alternative explanations, and we find little evidence that they are empirically important drivers of the policy impacts.

First, starting in June 2015, in its attempt to appease city legislators, Airbnb reported its willingness to remove some properties that “do not reflect our (Airbnb’s) vision for our community” in New York City and San Francisco.²¹ One may argue that the decreases in rents and home values were because of these periodic removals of properties by Airbnb instead of the OHOH policy. In other words, under this alternative explanation, the impact of the removals starting in June 2015 should dominate the reduction in rents and home values. One would only worry if the OHOH policy did not have any impacts after controlling this alternative event. To test if this is true, we add a dummy variable $1(\text{After2015m6})_{it}$ to indicate the alternative event for New York City and San Francisco in Equation (1). The results are presented in Table A2 in Section A2 in the online appendix. We do not find a significant impact on rents after Airbnb implemented the alleged removals. The removal event, which is similar to the OHOH policy, decreased the home values by displacing the supply from Airbnb to local markets. In either case, the two OHOH policy dummies continue to have negative, statistically significant coefficients, suggesting that the estimated impacts in Table 4 are indeed because of the OHOH policy rather than events such as those alleged removals.

Second, some may worry that other home-sharing platforms, such as HomeAway and VRBO (which are

Table 7. Policy Impacts on the Supply on Airbnb

	(1) log(ML properties)	(2) log(New properties)
1(Announcement) _{it}	−0.218*** (0.048)	−0.109 (0.101)
1(Implementation) _{it}	−0.164*** (0.038)	0.002 (0.089)
ACS controls	Yes	Yes
City-specific time trends	Yes	Yes
Zip code FE	Yes	Yes
Year-by-month FE	Yes	Yes
Observations	8,352	8,352
Adjusted R ²	0.839	0.776

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.

*** $p < 0.01$.

considered second and third to Airbnb),²² may follow Airbnb in imposing similar policies during the same time. In that case, our empirical findings will overestimate the impacts of the OHOH policy. We first conducted a diligent search for the news about these two competing platforms and found no similar policies in our study period. Also, a confounding policy that may completely invalidate our estimation of the policy impacts would have been rolled out in perfect overlap (in time and locations) with the OHOH policy. Such policies were unlikely. In addition, the likelihood of displacing properties to competing platforms, like HomeAway and VRBO, makes our estimated policy impacts (on local residential markets) conservative because these properties were not placed back in the local markets. Lastly, we check the number of properties in the policy-affected cities on HomeAway and VRBO (properties on other platforms are either overlapped or negligible compared with these two) and find that the market share of Airbnb is over 90% in all three policy-affected cities. The statistics are presented in Section A3 of the online appendix. Therefore, the intervening impact of other platforms, if any, might be too small to contaminate our estimates.

The third concern is that the removed Airbnb properties may have been relisted on Airbnb by hosts using new accounts under the names of friends or family. First, we argue that, although it is possible to do so, it might be too costly for hosts to manage multiple properties using other names that need to be verified (with a driver license number, taxpayer identification numbers, phone number, email account, etc.), making them financially and legally liable to the account holder (tax, fees, insurance, and legal responsibility if incidents occur). Second and more importantly, our estimates of the policy impacts would be conservative in this case, too. To see this, our evidence so far has suggested that the policy impacts on prices were driven by the increased supply in local residential markets. If hosts displaced the removed properties back to Airbnb using manipulated accounts, it would only imply a smaller increase in the supply in local residential markets (than the case in which the hosts do not list the removed properties back to Airbnb), which should lead to a smaller reduction in rents and home values. Therefore, our analyses would only underestimate the policy impacts if some hosts listed the removed properties back to the platform.

Nevertheless, we explore our Airbnb data set to examine the magnitudes of such strategic relisting behavior and whether it may bias our findings. Fortunately, the data set allows us to do so. Using the unique geo-coordinates of all properties (i.e., latitudes and longitudes), we look for properties at the same location but managed by different host identifications before and after the policy. Our exact matching of

latitudes and longitudes (i.e., at the most granular six-decimal level) identifies only 12 “relisted” properties. Of the total 180,264 properties in the three affected cities, this small subset accounts for only 0.006% of all policy-affected properties. We further relax the matching to a less granular level at five decimal points and identify 23 “relisted” properties, still accounting for only 0.01% of all properties in affected zip codes. Even if we continue to relax the matching to four decimal points, which allows two properties that are about 11 meters apart to be matched, there are only 2,054 “relisted” ones or 1.14% of all affected properties. We argue that these “suspiciously relisted” properties, given their negligible share in the market, are unlikely to attenuate our findings (let alone that some of them are not necessarily relisted properties and therefore, not outcomes of the strategic relisting behavior by hosts because of the policy restriction).

5.3. Further Analyses

5.3.1. Intensity of the Policy Impacts. We argue that studying the OHOH policy can be viewed as a useful way to date the onset of changes in local market equilibrium in response to policies adopted by home-sharing platforms or legislators. Despite the CEM matching algorithm and the extensive set of control variables included in our tests, one may still worry that our estimated policy impacts may not necessarily apply to other markets if similar policies were to be adopted. To overcome the challenge, we estimate the intensity of the policy impacts by exploring the moderating role of the density of policy-affected properties (i.e., ML properties). The main benefit of estimating the intensity of the policy impacts is that this method offers a scaled measure of the policy impacts. In addition, the results will provide reference points for legislators or other stakeholders if they attempt to regulate Airbnb in the markets they oversee.

The density of ML properties is the number of ML properties per 1,000 local population *before* the policy. The empirical specification accommodating the moderating effects is similar to our main specification (Equation (1)) and relies on the CEM sample. The regression equation is

$$Y_{it} = \beta_1 1(\text{Announcement})_{it} \times \log(\text{ML density} + 1)_i + \beta_2 1(\text{Implementation})_{it} \times \log(\text{ML density} + 1)_i + \beta_3 1(\text{Announcement})_{it} + \beta_4 1(\text{Implementation})_{it} + \gamma' Z_{it} + \mu_t + v_i + \varepsilon_{it}, \quad (3)$$

where $\log(\text{ML density} + 1)_i$ is the (logged) density of the policy-affected ML properties in a zip code i before the policy. Note that this measure is time invariant. The key coefficients, β_1 and β_2 , capture the intensity of the policy impacts. The percentage change

Table 8. Intensity of the Policy Impacts on Prices

Variables	(1) log(ZRI)	(2) log(ZHVI)	(3) log(Price2Rent)
$1(\text{Announcement})_{it} \times \log(\text{ML density} + 1)_i$	−0.010 (0.009)	0.013 (0.020)	0.024 (0.017)
$1(\text{Implementation})_{it} \times \log(\text{ML density} + 1)_i$	−0.045*** (0.015)	−0.046*** (0.016)	−0.002 (0.012)
$1(\text{Announcement})_{it}$	−0.011* (0.006)	−0.016*** (0.006)	−0.005 (0.007)
$1(\text{Implementation})_{it}$	−0.011 (0.007)	−0.006 (0.005)	0.005 (0.007)
ACS controls	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes
Observations	8,352	8,352	8,352
R^2	0.989	0.997	0.977

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.

* $p < 0.1$; *** $p < 0.01$.

in the dependent variables (prices and other local residential market metrics) can then be attributed to the policy if the density of ML properties in a zip code is 1% higher. We include the same set of control variables as in Equation (1).

Table 8 reports the results from estimating Equation (3), with the local residential market prices (log(ZRI), log(ZHVI), and log(Price2Rent)) being the dependent variables. We only report the coefficients of the interaction terms and the policy dummies for conciseness. In columns (1) and (2), the negative and statistically significant coefficients are consistent with our main results. To see this, we have shown that the policy effectively targeted ML hosts and affected local prices. We, therefore, expect the policy impacts to be more salient in locations (zip codes) with a higher density of ML properties because a larger portion of properties in those locations was subject to the policy restriction. We indeed find that a higher ML density in a local residential market is associated with larger price reductions after the policy. Second, the moderating effects of ML density are similar in the rental and housing markets, which is also consistent with our main results. The insignificant moderating effects on the price-to-rent ratio, again, confirm that the policy had similar impacts on the two local residential markets.

Numerically, our estimates suggest that the policy further decreased the rent by about 0.055% ($\approx -(0.010\% + 0.045\%)$) as in column (1) and the home value by roughly 0.033% ($\approx -(0.046\% - 0.013\%)$) as in column (2) if the density is 1% higher. In monetary terms, the estimates suggest that the policy helped save a typical renter an extra \$75 per year at the median annual rent in the policy-affected cities ($\approx$ median ZRI $\$1,134.6 \times 0.55\% \times 12$ months) and a typical home buyer an extra

\$803 at the median home value ($\approx$ median ZHVI $\$243,328.7 \times 0.33\%$) if the density of ML hosts increases by 10%. These estimates are extremely close to the estimates using the national medians. The median monthly rent of a one-bedroom apartment in the U.S. was about \$1,212 in December 2018,²³ and the median home value was roughly \$326,400 in December 2018.²⁴ On plugging in these national medians, our estimates suggest extra savings of \$80 in annual rents ($\approx \$1,212 \times 0.55\% \times 12$ months) and \$1,077 in home values ($\$326,400 \times 0.33\%$) if the density of ML properties is 10% higher.

Table 9 reports the moderating effects of the ML density on the supply in local residential markets. First, the positive and statistically significant coefficients of the interaction terms suggest that the policy led to greater increases in the supply in both markets where the density of ML properties was higher. This result is consistent with our findings of the policy impacts on the supply in local markets. This is also consistent with the estimated intensity of the policy impacts on prices (Table 8) because the policy led to even higher supply and hence, even lower prices in markets with a higher density of ML properties. Quantitatively, the estimates in columns (1) and (2) indicate that a 1% increase in the density of ML properties further magnifies the policy impact on the supply in housing markets by about 0.24% and the rental supply by 0.023%.

5.3.2. By-City Analyses. We further contemplate the generalizability of our results by evaluating the policy impacts and the intensity of the policy impacts in three affected cities separately. Because our CEM method matches zip codes in the same strata (instead of being one-to-one matching), we rerun the matching for the zip codes in each affected city separately. In

Table 9. Intensity of the Policy Impacts on Rental and Housing Supply

Variables	(1) log(<i>Housing supply</i>)	(2) log(<i>Rental supply</i>)
1(After policy) _{it} × log(<i>ML density</i> + 1) _i	0.237*** (0.071)	0.023*** (0.007)
1(After policy) _{it}	0.056** (0.027)	−0.002 (0.004)
ACS controls	Yes	Yes
City-specific time trends	Yes	Yes
Zip code FE	Yes	Yes
Year-by-month FE	Yes	Yes
Observations	4,992	8,352
R ²	0.522	0.718

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.
p* < 0.05; *p* < 0.01.

the end, we have 90, 12, and 26 matched zip codes in New York City, San Francisco, and Portland, respectively. In the matching process, we keep the same number of zip codes in the unaffected groups as in the matched affected groups.

We first report the by-city policy impacts on price indices in Table 10. Panel I presents the results on rents. In each regression, we use the matched zip codes in one affected city along with the matched unaffected zip codes. For example, the sample used in the regression of column (1), the “New York City” regression, contains 90 zip codes in New York City and 90 matched zip codes from across the country (excluding San Francisco and Portland). Note that because Airbnb implemented the policy in Portland without announcement, the “Portland” regression has only the implementation dummy. We can see that all policy dummies in three regressions are negative and statistically significant, suggesting that the policy

decreased the rents in all three cities. The magnitudes, however, vary. The policy decreased the rents in New York City by about 2.6%, in San Francisco by more than 12%, and in Portland by 5.9%. Compared with these by-city results, our main findings from combining the three cities present a relatively conservative estimate (Table 4). We find a similar pattern in by-city regressions on the policy impacts on home values, as reported in Panel II of Table 10. The policy also decreased home values in all three cities to varying degrees.

What could have caused the larger policy impacts in San Francisco? As the results in the previous subsection show, the policy impacts were higher in zip codes with a higher density of policy-affected properties (from ML hosts). We, therefore, compare the ML density in the matched zip codes across the three cities. The results are reported in Table 11. It is evident that San Francisco had a higher average density of ML

Table 10. Policy Impacts on Rents and Home Values by City

	Panel I: log(<i>ZRI</i>)			Panel II: log(<i>ZHVI</i>)		
	(1) New York City	(2) San Francisco	(3) Portland	(4) New York City	(5) San Francisco	(6) Portland
1(Announcement) _{it}	−0.014*** (0.004)	−0.046*** (0.012)		−0.007 (0.005)	−0.032** (0.014)	
1(Implementation) _{it}	−0.012** (0.006)	−0.078*** (0.018)	−0.059*** (0.011)	−0.011** (0.006)	−0.045** (0.016)	−0.041*** (0.010)
ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,528	1,152	2,784	6,528	1,152	2,784
Adjusted R ²	0.985	0.980	0.977	0.996	0.996	0.995

Notes. Because Airbnb implemented the policy in Portland directly, Portland has only one policy dummy, 1(Implementation). Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.
p* < 0.05; *p* < 0.01.

Table 11. Comparing the Density of ML Properties in Three Affected Cities

City	Number of ML properties per 1,000 population				
	Min	Max	Mean	Median	Standard deviation
San Francisco	0.096	1.904	0.837	0.836	0.573
New York City	0.000	3.340	0.363	0.097	0.660
Portland	0.000	2.319	0.585	0.307	0.638

properties (0.837), whereas the average ML properties per 1,000 population in New York City and Portland were 0.442 and 0.585, respectively, in their matched zip codes. The relatively higher density of ML properties in San Francisco may have caused the higher policy impacts.

Motivated by the difference in ML density, we continue to test whether the intensity of the policy impacts was relatively similar across the three cities. To do this, we replicate the intensity analysis in the last subsection for the three cities separately. Table 12 reports the results. We can see that the marginal impact of 1% higher ML density on rents is about 0.063% in New York City, 0.079% in San Francisco, and 0.048% in Portland (Panel I). As conjectured, the magnitudes are quite comparable across the three cities. We report the results on home values in Panel II. The estimates for New York City (0.049%), San Francisco (0.056%), and Portland (0.06%) are even closer to one another. Despite using different samples with different cities, these results, combined with our main findings of the intensity of the policy impacts (Table 8), illustrate that the policy had highly similar impacts across different locations in terms of intensity. We

regard these estimates as benchmarks for other cities that may consider implementing similar restrictions.

5.4. Robustness Checks

5.4.1. Extra Controls of the Supply and Demand in Local Residential Markets. Our matching method relies on a rich set of variables on local residential price indices, the market size of Airbnb, and local economic and demographic characteristics. Using the CEM-generated sample, we include an extensive set of variables to control for time-variant and time-invariant factors. One may argue that the supply and demand in local residential markets may have an impact on the prices. To control for the supply in local residential markets, we add the (logged) number of for-sale housing units (in a zip code in a month) on Zillow. Panel I of Table 13 reports the policy dummy coefficients and the housing supply coefficients. The estimates of the policy impacts controlling the housing supply continue to be qualitatively the same as our main findings.

To control for the demand in local residential markets, we use the average Google trends in local residents' search for housing-related keywords.²⁵ Panel II

Table 12. Intensity of Policy Impacts on Rents and Home Values by City

	Panel I: log(ZRI)			Panel II: log(ZHVI)		
	(1) New York City	(2) San Francisco	(3) Portland	(4) New York City	(5) San Francisco	(6) Portland
1(Announcement) _{it} × log(ML density) _i	−0.021** (0.010)	−0.062** (0.025)		0.007 (0.012)	0.033 (0.030)	
1(Implementation) _{it} × log(ML density) _i	−0.042*** (0.010)	−0.017 (0.021)	−0.048** (0.021)	−0.056*** (0.009)	−0.089*** (0.016)	−0.060** (0.023)
1(Announcement) _{it}	−0.009* (0.005)	−0.015 (0.020)		−0.009* (0.005)	−0.052*** (0.018)	
1(Implementation) _{it}	−0.002 (0.006)	−0.065*** (0.020)	−0.042*** (0.013)	0.002 (0.005)	0.006 (0.017)	−0.020* (0.012)
ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,528	1,152	2,784	6,528	1,152	2,784
Adjusted R ²	0.986	0.981	0.978	0.996	0.996	0.995

Notes. Because Airbnb implemented the policy in Portland directly, Portland has only one policy dummy, 1(Implementation). Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 13. The Policy Impacts by Controlling Housing Supply or Google Trends

	Panel I: Controlling housing supply			Panel II: Controlling Google trends		
	(1) log(ZRI)	(2) log(ZHVI)	(3) log(Price2Rent)	(4) log(ZRI)	(5) log(ZHVI)	(6) log(Price2Rent)
1(Announcement) _{it}	−0.013** (0.005)	−0.020*** (0.006)	−0.008 (0.008)	−0.011*** (0.003)	−0.018*** (0.006)	−0.007 (0.008)
1(Implementation) _{it}	−0.023*** (0.007)	−0.016*** (0.005)	0.007 (0.007)	−0.008*** (0.003)	−0.013*** (0.005)	−0.005 (0.007)
log(Housing supply)	0.002 (0.007)	0.003 (0.006)	0.001 (0.008)			
Average Google trends				0.001*** (0.000)	0.000** (0.000)	−0.000 (0.000)
ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,992	4,992	4,992	6,912	6,912	6,912
R ²	0.988	0.997	0.982	0.988	0.997	0.978

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.
p* < 0.05; *p* < 0.01.

of Table 13 presents the results with the extra control, which remain qualitatively the same as our main findings as well.

5.4.2. Alternative Specifications.

5.4.2.1. DID with Propensity Score Matching. One may worry that our main findings (e.g., Tables 5, 7, and 8) are driven by the matching method we choose (i.e., the CEM method). To alleviate the concern, we use an alternative propensity score matching (PSM) method. The PSM process generates 151 matched zip codes in the affected group. We further keep only the nearest neighbor for each of the affected zip codes, resulting in 151 zip codes forming the unaffected group. As shown in Table 14, the mean values of all

covariates before the policy in the affected and unaffected groups are close to one another, with none of the pairs having a statistically significant difference.

Table 15 reports the DID estimates of the policy impacts on price indices using the PSM sample. Although the magnitudes of the estimates are slightly different for rents and home values, we find consistently negative impacts on both price indices. Also, the combined estimated impacts, 3.4% and 2.2%, respectively, are not far from our main findings in Table 4.

5.4.2.2. A Synthetic Control Method. We utilize a synthetic control method to further alleviate the concern of comparability. This method involves, for each

Table 14. Characteristics of Unaffected Zip Codes Vs. Affected Zip Codes in the PSM Sample

	Unaffected zip codes			Affected zip codes			<i>t</i> test	
	Mean	Standard deviation	<i>N</i>	Mean	Standard deviation	<i>N</i>	Difference	<i>p</i> -value
Rental and housing market price indices								
log(ZRI)	0.268	0.412	151	0.228	0.344	151	−0.041	0.352
log(ZHVI)	5.710	0.680	151	5.637	0.563	151	−0.073	0.309
log(Price2Rent)	2.957	0.304	151	2.925	0.283	151	−0.032	0.338
The market size of Airbnb								
log(Airbnb Properties)	3.802	1.390	151	3.633	1.611	151	−0.169	0.333
Economic and demographic characteristics								
log(Population)	10.286	0.691	151	10.271	0.814	151	−0.015	0.860
log(Households)	9.336	0.667	151	9.321	0.744	151	−0.015	0.851
log(Med. income)	11.106	0.385	151	11.080	0.380	151	−0.026	0.562
Unemployment rate	7.993	2.700	151	8.394	3.027	151	0.401	0.225
Home vacancy rate	8.765	6.362	151	8.991	6.224	151	0.226	0.755
% White	44.845	21.611	151	42.055	24.235	151	−2.789	0.292
% Hispanic	14.809	13.152	151	14.747	11.508	151	−0.061	0.966
% Female	50.817	2.418	151	51.152	3.125	151	0.335	0.298

Table 15. Policy Impacts on Local Residential Market Prices Using the PSM Sample

	(1) log(ZRI)	(2) log(ZHVI)	(3) log(Price2Rent)
1(Announcement) _{it}	−0.014*** (0.004)	−0.010* (0.005)	0.004 (0.005)
1(Implementation) _{it}	−0.020*** (0.005)	−0.012*** (0.004)	0.008 (0.005)
ACS controls	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes
Observations	14,466	14,466	14,466
R ²	0.994	0.997	0.982

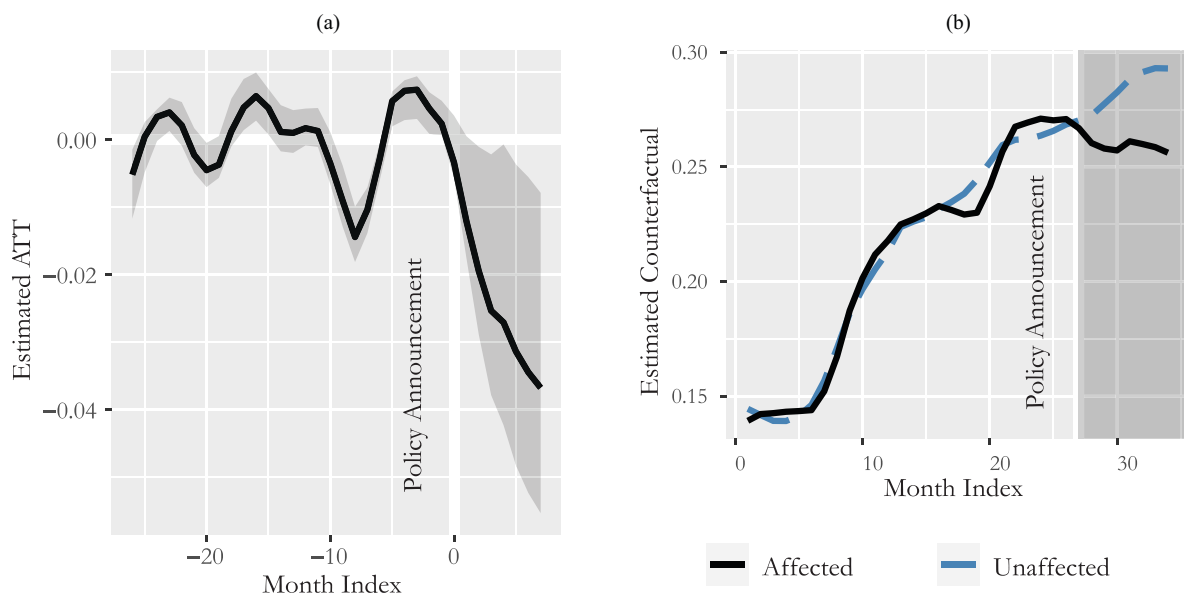
Notes. Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.

* $p < 0.1$; *** $p < 0.01$.

affected zip code, the construction of a weighted combination of zip codes in the unaffected group to form a “synthetic control” zip code that can best mirror the affected zip code (Krijestorac et al. 2020). Then, the constructed “control” zip codes are used to estimate what would have happened to the affected zip codes if they were not affected by the policy (Athey and Imbens 2017). Compared with the usual DID approach that essentially takes a simple average of all zip codes in unaffected cities, an advantage of the synthetic control method is that it accounts for the effects of confounders changing over time by weighting the “control group” to better match the “treatment group” before the policy (Abadie et al. 2010). Because

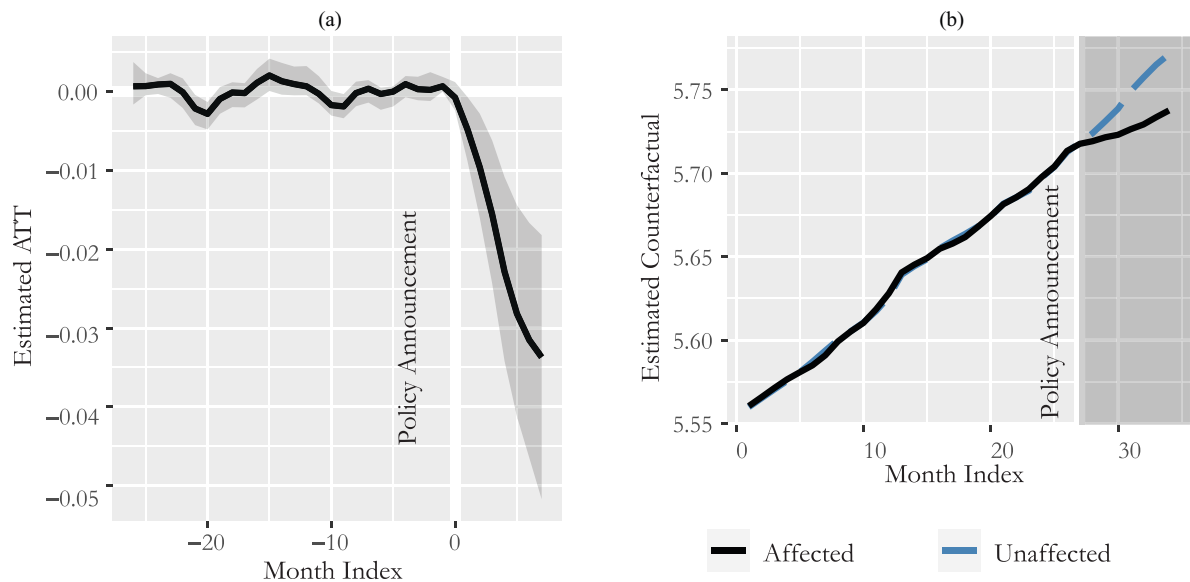
we have multiple policy-affected zip codes, we rely on a variant of the traditional synthetic control method, a generalized synthetic control (GSC) method developed in Xu (2017), which allows for multiple “treated” units and estimates the “average treatment effects” on these “treated” units based on an interactive fixed effects model (Bai 2009).

Figure 3 presents the findings of the policy impact on rents from the GSC method. We use only the policy announcement that happened in New York City and San Francisco in April 2016 for a clear demonstration (based on the CEM sample). Panel (a) illustrates the estimated coefficient of the policy announcement, where the x axis is the relative months to the announcement (with zero being the announcement month) and the y axis represents the difference between the affected and unaffected zip codes. The grey area in panel (a) is the 95% confidence interval. As panel (a) shows, the difference between the affected zip codes and the synthetic control zip codes centers around zero before the policy. In other words, the two groups are similar in terms of rents before the announcement. However, after the policy announcement, the rents in the affected zip codes started to decrease significantly compared with those in the synthetic control counterfactuals. Panel (b) illustrates the different policy impacts between the affected and unaffected zip codes. We can see that the solid line, representing the affected zip codes, overlaps with the dotted line (for the synthetic unaffected zip codes) before the policy announcement but falls below the

Figure 3. (Color online) Impact of the Policy on Rents Using a Synthetic Control Method

Notes. (a) The estimated policy impact on rents. (b) The “treated” vs. the synthetic control. Panel (a) draws the gap plot—the average treatment effect on the treated (ATT) of the policy. Panel (b) depicts the counterfactuals for affected zip codes. We use only policy announcements that happened in New York City and San Francisco at the same time for a clear demonstration.

Figure 4. (Color online) Impact of the Policy on Home Values Using a Synthetic Control Method



Notes. (a) The estimated policy impact on home values. (b) The “treated” vs. the synthetic control. Panel (a) draws the gap plot—the average treatment effect on the treated (ATT) of the policy. Panel (b) depicts the counterfactuals for affected zip codes and others separately. We use only policy announcements that happened in New York City and San Francisco at the same time for a clear demonstration.

latter afterward, which reaffirms that the policy led to a reduction in rents in affected zip codes. Figure 4 shows the graph for the policy impact on home values based on the GSC method. We can see a similar pattern that verifies our main findings.

5.4.3. Ashenfelter’s Dip. Although the results from our relative time model show no difference in the pre-policy trend between affected and unaffected zip codes, one may still worry that the existence of any difference in such trends may bias our estimates, which is commonly termed Ashenfelter’s Dip in the DID literature (Ashenfelter and Card 1985). We follow the common practice to correct for Ashenfelter’s Dip by excluding the sample periods around the policy

shock. Specifically, we exclude (1) the observations three months before and three months after the policy announcement (i.e., January to June 2016) and (2) the observations three months before the announcement (January to March 2016). By removing the prepolicy trends, the decrease in rents and home values should be smaller. The results are reported in Table 16. The coefficients are indeed smaller in magnitude compared with the main estimates in Table 4 but are still qualitatively the same. Both rents and home values decreased after the policy, and the change in the price-to-rent ratio was not significant.

5.4.4. A Placebo Test of Random Assignment. One may argue that there happened to be a decrease in

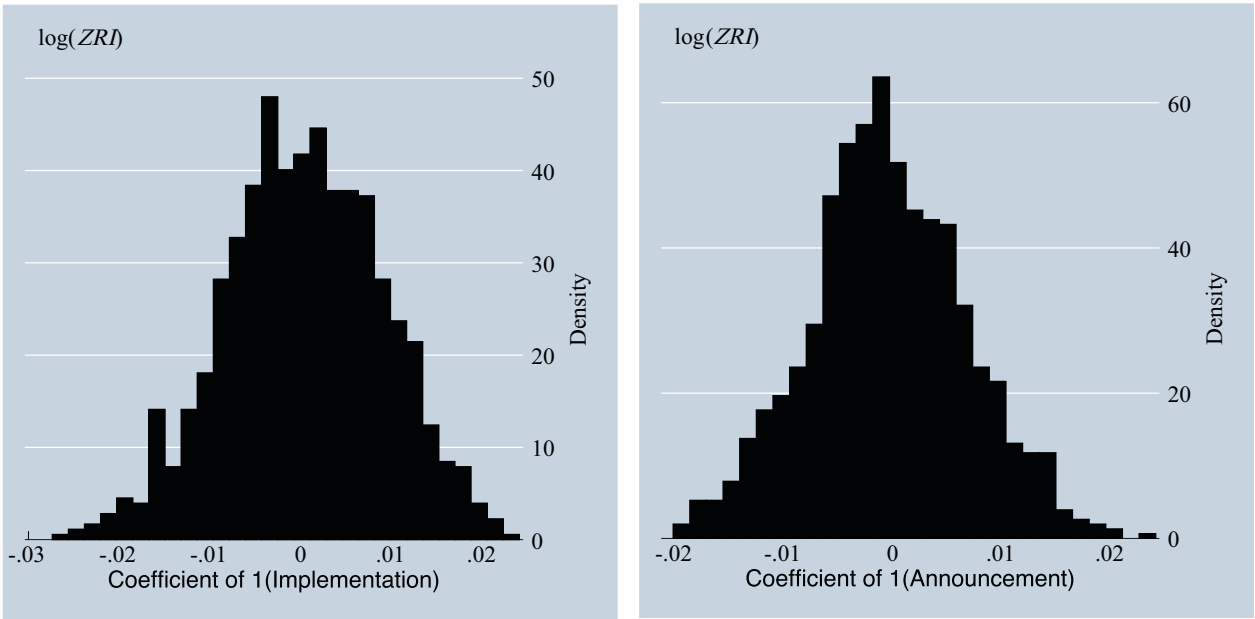
Table 16. DID Results by Excluding Three Months Prior to the Policy Announcement

	(1) log(ZRI)	(2) log(ZHVI)	(3) log(Price2Rent)	(4) log(ZRI)	(5) log(ZHVI)	(6) log(Price2Rent)
1(Announcement) _{it}	−0.008 (0.006)	−0.019*** (0.007)	−0.011 (0.009)	−0.011 (0.008)	−0.021** (0.008)	−0.010 (0.011)
1(Implementation) _{it}	−0.018*** (0.007)	−0.016*** (0.005)	0.002 (0.006)	−0.016*** (0.006)	−0.013*** (0.004)	0.003 (0.006)
ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
City-specific time trends	Yes	Yes	Yes	Yes	Yes	Yes
Zip code FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,830	7,830	7,830	7,308	7,308	7,308
R ²	0.987	0.997	0.978	0.987	0.997	0.978

Notes. We drop three months right before policy announcement in columns (1)–(3) and three months both before and after policy announcement in columns (4)–(6). Robust standard errors clustered at the zip code level are shown in parentheses. FE stands for fixed effect.

p* < 0.05; *p* < 0.01.

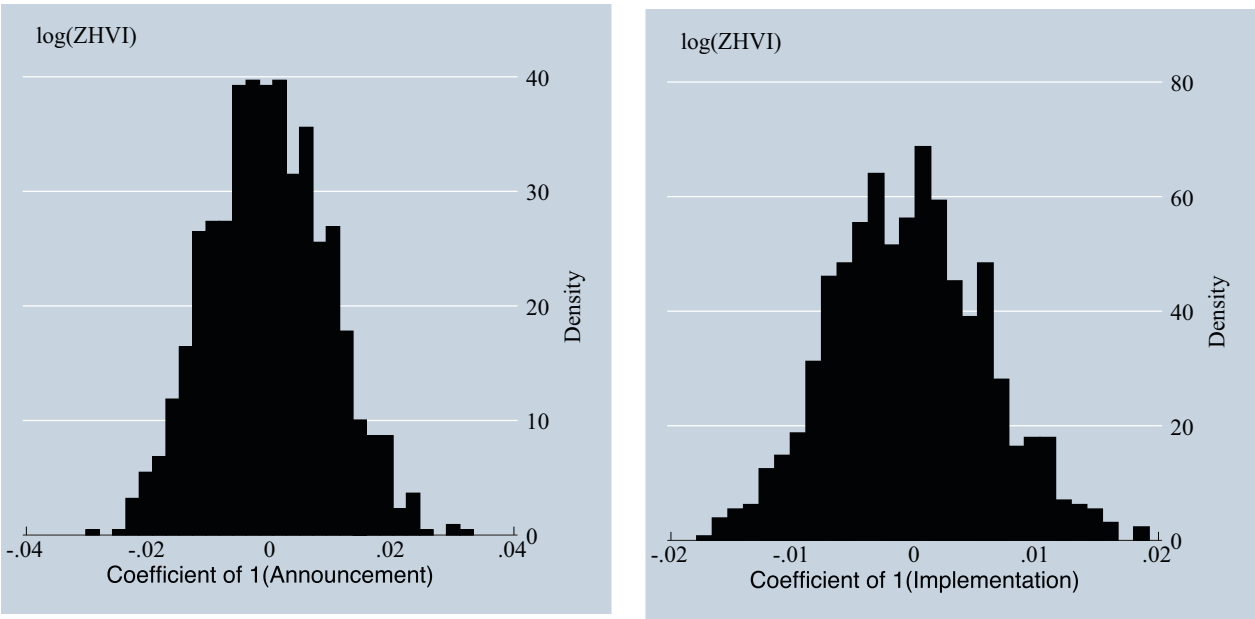
Figure 5. (Color online) A Placebo Test of Randomized Assignment: Rents



both rents and home values in the policy-affected markets relative to unaffected markets regardless of the OHOH policy. To ensure that it is the policy driving the results, we conduct a placebo test in which we randomly assign a subset of unaffected zip codes as if they were affected by the policy. Specifically, we focus on only unaffected zip codes in this test. In each draw, we randomly treat 50% of those zip codes as if they

were subject to the OHOH policy and replicate our main regressions to obtain the coefficients of the two policy dummies. We repeat this process multiple times in our simulations. Figures 5 and 6 depict the distributions of the estimated coefficients from simulations with 1,000 draws each. In both figures, panel (a) presents the coefficient for the announcement dummy, and panel (b) displays the coefficient for the

Figure 6. (Color online) A Placebo Test of Randomized Assignment: Home Values



implementation dummy. As we can see from the figures, in all four cases, most of the mass (of the distributions) centers around zero between -0.01 and 0.01 . This suggests that all the estimated coefficients are not significantly different from zero (in a statistical sense), implying that the “fake” policies constructed in the simulations have no effect on either rents or home values. The results from this placebo test lend further support to our findings that it is the policy driving the differences in local residential market prices (between affected and unaffected zip codes).

6. Concluding Remarks

The progress in information technologies has given rise to a host of online marketplaces that facilitate transactions between buyers and sellers without traditional intermediations, such as real estate brokers or financial agents. The lower transaction costs and search costs provide these platforms with advantages over incumbent channels (Goldfarb and Tucker 2019). Such advantages are especially salient in real estate residential markets, in which home-sharing services, such as Airbnb, arise with the proliferation of internet-related technologies. Suppliers in local residential markets, be they homeowners or absentee landlords, now have an alternative channel to allocate assets. Yet, the impacts of home-sharing platforms on local residential markets are largely underexplored. Because the emergence of these platforms leads to a natural concern that they might harm housing affordability, understanding the impact on local residential markets and the underlying mechanism is important both theoretically and practically.

We respond to the call by answering whether and how the home-sharing platforms make rentals and housing less affordable. On the supply side of local residential markets, home-sharing platforms have evolved as a major alternative channel for property owners to rent out their accommodation space. In contrast, on the demand side, few local residents would resort to home sharing for long-term accommodation. We demonstrate this unique dynamic between home sharing and local residential markets using Airbnb’s “One Host, One Home” policy and find that the policy drove down both rents and home values in local residential markets. We further drill down to the mechanism that the policy led to increased supply in rental and housing markets—namely, the policy forced absentee landlords to put their properties back in local residential markets and retained future entries to comply with the “One Host, One Home” restriction. This mechanism prevails even after we account for a few intervening events. These findings speak directly to how home sharing disrupts local residential markets, i.e., as a major substitute for the supply

in local markets (rather than the demand in local residential markets). Lastly, we estimate the intensity of the policy impacts and find that the policy may further decrease home values and rents by about 0.3% – 0.6% if the density of affected Airbnb properties is 10% higher in a market, similar for each affected city. One advantage of our study is that we utilize a handful of empirical methods (to cross-validate our findings) that include a DID combined with a CEM approach as the main specification, a relative time model, a DID combined with a PSM method, and a state-of-the-art synthetic control method that allows for multiple policy-affected zip codes. Our main results survive multiple alternative specifications, robustness checks, and placebo tests, consistently pointing to similar findings.

We are among a few pioneers to explore the dynamics between home sharing and local residential markets (Sheppard and Udell 2016, Horn and Merante 2017, Barron et al. 2021). Specifically, using the unique platform policy that caps the properties a host can manage on Airbnb, our findings point out that the mechanism of home sharing disrupting local markets centers on the displacement of residential properties to the online channel. Our findings suggest that the extent of the displacement has had a major impact on the supply in local residential markets, translating into the price levels (rents and home values) and hence the housing affordability concerns. We are among the first to study how a policy initiated and implemented by a technology platform could affect a major index of an economy (i.e., housing affordability). Existing studies predominantly focus on government policies (e.g., Valentin 2021). Lastly, we show that technology platforms can self-govern and that the technology-facilitated policy is observed to be highly effective. As platforms continue to rise and scale, we suggest that stakeholders of the platforms are mindful of unexpected societal impacts (e.g., housing affordability issues studied in this paper) and proactively self-govern for goodwill.

Acknowledgments

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Endnotes

¹ Properties listed on home-sharing platforms are primarily rented by transient travelers for a relatively short period of time, typically less than a week. In sharp contrast, local residents seek rentals for a much longer period—from a few months to one or two years (Barron et al. 2021).

² We consider the residential real estate market as two relatively separate markets—the long-term rental market and the rest, which is a market for homes for sale by either homeowners or real estate agents. The literature does not have a general name for this market. Examples are housing markets (Sommer and Sullivan 2018) and for-sale markets (Barron et al. 2021). To fix terminology, we call it the “housing market” throughout the manuscript. In terms of the market prices, we unanimously use “rents” to indicate the periodic (e.g., monthly or annual) payments from tenants to landlords and use “home values” to refer to the prices paid to purchase a home.

³ Examples are ample, such as London and Amsterdam in Europe; Hong Kong and Singapore in Asia; and Los Angeles, Chicago, San Francisco, and Seattle in the United States. The regulations enforced by these cities are different, but they all appear to focus on, by and large, absentee landlords; some limit the rental property to the host's primary residence, some only allow a property up to 90 nights a year without hosts, and others require hosts with multiple properties to obtain a license from the city. Short-term rental laws in major U.S. cities can be found at <https://www.2ndaddress.com/research/short-term-rental-laws> (retrieved on January 29, 2022). In some extreme cases like Berlin, the local government banned home sharing with the support of a court ruling (O'Sullivan 2016).

⁴ Airbnb's official site provides a detailed description of the policy (<https://web.archive.org/web/20180313124259/https://www.airbnb.com/help/article/1333/why-can-t-i-have-listings-at-more-than-one-address-in-select-locations>); retrieved on January 29, 2022. The rolling out to Portland can be found here: <https://katu.com/news/local/airbnb-launches-one-host-one-home-policy-in-portland-to-protect-long-term-housing> (retrieved on January 29, 2022). Around the same time, Airbnb experimented with this policy outside the United States in several areas in Barcelona, Spain. Up to the end of the study period, the cities in the United States where the OHOH policy was rolled out only include New York City, San Francisco, and Portland, all considered in this study.

⁵ Hosts with legitimate reasons to have listings at different addresses (e.g., traditional bed and breakfasts, boutique hotels, long-term corporate housing, long-term rentals (for 30+ nights only), and licensed hotels (or similar)) were exempt from the policy (Airbnb 2016). Such exemptions do not invalidate our empirical findings. First, the market share of such listings on Airbnb is negligible. More importantly, the existence of such properties makes our estimates of the policy impacts (and in turn, the impact of home sharing on housing affordability) conservative because these properties can still be displaced from the local residential markets to Airbnb under the policy.

⁶ Our CEM procedure drops some zip codes in the policy-affected cities because of the difficulty in finding similar markets elsewhere. For example, some zip codes in the Manhattan area of New York City are dropped because of their high population density and high home values, as detailed in Section 4. In addition to the CEM

sample, we test the robustness of our findings by constructing another sample using PSM. The results from the two samples (the CEM and PSM samples) are qualitatively consistent.

⁷ The median ZRI (in 1990 US dollars) of the policy-affected zip codes in our sample is about \$1,134.6, and the median ZHVI in our sample is about \$243,328.7. Based on our estimates, the reduction in annual rents was roughly $\$1,134.6 \times 0.031 \times 12 \approx \422 , and the reduction in home values was about $\$243,328.7 \times 0.029 \approx \$7,057$. We apply similar calculations to the policy impacts on rents and home values in the remaining paper.

⁸ In our definition, the density of ML properties is the number of ML properties per 1,000 local population (detailed in Section 5), which is different from population density measures that are typically the population per square mile (or per square kilometers). Therefore, a market with a high density of ML properties does not necessarily equate to a market of high population density.

⁹ Barron et al. (2021) propose a Bartik-type instrument that currently sees extensive conversations in the economics literature (see, e.g., Pinkham et al. 2020). The initial form of the instrument is defined as “the local employment growth rate predicted by interacting local industry employment shares with national industry employment growth rates.” Under certain settings (where researchers have a preperiod), an empirical strategy that utilizes Bartik-type instruments can be viewed as a DID design (Pinkham et al. 2020).

¹⁰ The Consumerist documents a growing concern about the so-called “mega hosts” who are profit seekers, renting out dozens, hundreds, sometimes more than 1,000 properties on Airbnb (source: <https://consumerist.com/2017/06/09/some-airbnb-mega-hosts-are-renting-out-more-than-1000-properties-at-once/>; retrieved on January 29, 2022).

¹¹ Airbnb provides more information about the policy on its web page: <https://web.archive.org/web/20180313124259/https://www.airbnb.com/help/article/1333/why-can-t-i-have-listings-at-more-than-one-address-in-select-locations> (retrieved on January 29, 2022).

¹² The results from our relative time models, reported in Section 5, lend support and demonstrate that there were unlikely expectations about when, where, or how a similar type of policy may roll out.

¹³ Introductions to VACASA (<https://www.vacasa.com/>), CBRE Group (<https://www.cbre.us/>), Los Angeles Tourism & Convention Board (<https://www.discoverlosangeles.com/>), and Bank of America Merrill Lynch (<https://www.bofam.com>) can be found via their respective links.

¹⁴ To protect privacy and security, Airbnb does not display the exact address on the property page. Customers can only get an estimated location within a small radius when browsing the listing page. The full address of a listing will only be given after it is booked.

¹⁵ Zillow's methodology of calculating the price indices, ZRI and ZHVI, can be found on their data export web page at <https://www.zillow.com/research/data/>. Note that Zillow renamed the ZRI as Zillow Observed Rent Index or ZORI.

¹⁶ Not surprisingly, translating the price indices into 2000 or 2010 U.S. dollars yields similar results in our empirical analysis.

¹⁷ Alternatively, we could present these two variables as interaction terms between a policy-affected dummy, indicating a policy-affected zip code, and two event dummies, indicating that the period is after the announcement or implementation, respectively. Note that under this alternative specification, the individual terms—the policy-affected dummy and the two event dummies—are not identified because we include both zip code and year-by-month dummies.

¹⁸ In the following discussions, even though we still present the effects of the policy announcement and implementation separately

in the results tables, we often combine the two effects to discuss the overall impact of the policy.

¹⁹ Only the time dummy indicating at least six months before the announcement in column (1) is significant at a 10% level.

²⁰ The decreased new entries also alleviate the concern that hosts may strategically bypass the policy by cannibalizing their multilistings into single listings and putting them back on the platform.

²¹ See the reports on Airbnb's removal starting June 2015 in New York City at <https://www.airbnbncitizen.com/wp-content/uploads/2016/07/OneHostOneHomeNewYorkCity-1.pdf> (retrieved on January 29, 2022) and in San Francisco at <https://www.businessinsider.com/airbnb-vows-to-crack-down-on-illegal-hotels-in-san-francisco-2016-4> (retrieved on January 29, 2022).

²² VRBO and HomeAway—both owned by Expedia Group and displaying identical properties—are the second and third most popular home-sharing services (source: <https://www.airgms.com/vacation-rental-sites/>; retrieved on January 29, 2022). Other platforms, such as Booking.com and TripAdvisor.com, also offer home-sharing services yet on a much smaller scale. Marriott reportedly plans to create a platform for home sharing, which may be a viable competitor to Airbnb in the future (source: <https://www.wired.com/story/airbnb-marriott-each-want-what-other-has/>; retrieved on January 29, 2022).

²³ The source is <https://www.zumper.com/blog/2018/11/zumper-national-rent-report-december-2018/> (retrieved on January 29, 2022).

²⁴ The source is https://www.census.gov/construction/nrs/historical_data/index.html (retrieved on January 29, 2022).

²⁵ Examples of the keywords include “housing,” “rents,” and “home values.” A caveat is that Google only provides historical trend data at the city level. Recall that our main unit of analysis is zip code by month. Therefore, we match the Google trend data to the city where a zip code belongs.

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